advocated and should be discontinued if no benefit is observed. $H_2$ antagonists also improve symptoms in functional dyspepsia, but a guideline advocated PPIs over $H_2$ antagonists as first-line therapy.

Antacids are useful for short-term control of mild GERD but have less benefit in severe cases unless given at high doses that cause side effects (diarrhea and constipation with magnesium- and aluminum-containing agents, respectively). Alginic acid combined with antacids forms a floating barrier to reflux in patients with postprandial or nocturnal symptoms. Sucralfate, a salt of aluminum hydroxide and sucrose octasulfate that buffers acid and binds pepsin and bile salts, shows efficacy in GERD similar to $H_2$ antagonists.

### *HELICOBACTER PYLORI* ERADICATION

The benefits of *H. pylori* eradication in functional dyspepsia are limited but are statistically significant with a 9% risk reduction compared to placebo. A systematic review of 25 controlled trials calculated a pooled risk ratio of 1.24, with a 95% confidence interval of 1.12–1.37, favoring *H. pylori* eradication over placebo. Most drug combinations **(Chaps. 168 and 335)** include 7–14 days of a PPI plus two or three antibiotics with or without bismuth products. *H. pylori* infection is associated with reduced prevalence of GERD. However, *H. pylori* eradication does not worsen GERD symptoms.

### AGENTS THAT MODIFY GASTROINTESTINAL MOTOR ACTIVITY

The γ-aminobutyric acid B (GABA-B) agonist baclofen reduces esophageal exposure to refluxed gastric contents by reducing TLESRs by 40% and can be used in patients with refractory regurgitation of acid or nonacid fluid. Several studies promote the efficacy of agents that stimulate gastric emptying in functional dyspepsia with 33% relative risk reductions, but publication bias and small sample sizes raise questions about the true benefits of these agents. The PDS subtype may respond preferentially to such prokinetic drugs. Although the 5-$HT_4$ agonist prucalopride can reduce symptoms in idiopathic gastroparesis, no similar studies have been conducted in functional dyspepsia. The 5-$HT_{1A}$ agonists buspirone and tandospirone may improve symptoms in functional dyspepsia and gastroparesis by enhancing meal-induced gastric accommodation. Acotiamide stimulates gastric emptying and augments accommodation by enhancing acetylcholine release via muscarinic receptor antagonism and acetylcholinesterase inhibition. This agent is approved for functional dyspepsia in Japan and India.

### CENTRALLY ACTING NEUROMODULATORS

Some patients with refractory functional heartburn respond to tricyclic antidepressants. Their mechanism of action may involve blunting of visceral pain processing in the brain. In one controlled trial in functional dyspepsia, the tricyclic drug amitriptyline produced symptom reductions, whereas the selective serotonin reuptake inhibitor (SSRI) escitalopram had no benefit in a three-way comparison with placebo. A second study reported greater improvement with a different tricyclic agent imipramine versus placebo. In a controlled trial of a different antidepressant, mirtazapine produced superior symptom reductions in functional dyspepsia compared to placebo. In contrast, a meta-analysis of 13 trials of SSRIs and serotonin-norepinephrine reuptake inhibitors observed no benefits in this disorder.

### OTHER OPTIONS

Gas and bloating may be difficult to control in patients with indigestion. Simethicone, activated charcoal, α-galactosidase, and other over-the-counter herbal products provide benefits in some cases. One trial suggested possible benefits of the nonabsorbable antibiotic rifaximin in functional dyspepsia, while another reported improvement with the probiotic *Lactobacillus gasseri*. Herbal remedies like STW 5 (a mixture of nine herbal agents) and formulations of caraway oil and menthol show efficacy in some dyspeptic patients in controlled studies. Acupuncture has shown benefits in sham-controlled trials. Psychological treatments (e.g., behavioral therapy, psychotherapy, hypnotherapy, diaphragmatic breathing, mindfulness) may be offered for refractory heartburn due to esophageal hypersensitivity or functional dyspepsia; a meta-analysis of four trials reported benefits in patients with persistent dyspepsia.

Antireflux surgery (fundoplication) to enhance the barrier function of the LES may be offered to GERD patients who are young and require lifelong therapy, have typical heartburn, are responsive to PPIs, and show acid reflux on pH monitoring. Surgery also is effective for some cases with prominent regurgitation. Factors associated with poor response to fundoplication include atypical reflux symptoms, no pathologic acid reflux on pH testing, esophageal body motor disturbances on manometry, and delayed gastric emptying. Dysphagia, gas-bloat syndrome, and gastroparesis are long-term complications of fundoplication; ~60% develop recurrent GERD symptoms over time. Magnetic sphincter augmentation may be appropriate for GERD treatment, while endoscopic transoral incisionless fundoplication can be considered for some patients.

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# 49 Diarrhea and Constipation

**Michael Camilleri, Joseph A. Murray**

Diarrhea and constipation are exceedingly common and, together, exact an enormous toll in terms of mortality, morbidity, social inconvenience, loss of work productivity, and consumption of medical resources. Worldwide, >1 billion individuals suffer one or more episodes of acute diarrhea each year. Among the 100 million persons affected annually by acute diarrhea in the United States, nearly half must restrict activities, 10% consult physicians, ~250,000 require hospitalization, and ~5000 die (primarily the elderly). Updated 2021 annual disease burden data from the United States show 4.05 million annual clinic or emergency department visits for constipation or diarrhea or irritable bowel syndrome (IBS), and annual median economic burden to society for functional intestinal disorders of $145 million. Acute infectious diarrhea is the second leading cause of death in children, and the World Health Organization reports around 525,000 children under age 5 years die each year from acute diarrhea. Recurrent, acute diarrhea in children in tropical countries results in environmental enteropathy with long-term impacts on physical (stunted growth) and intellectual development.

Constipation, by contrast, is rarely associated with mortality and is exceedingly common in developed countries, leading to frequent self-medication and, in a third of those, to medical consultation. Annual disease burden data for 2016 show about 5 million clinic or emergency department (ED) visits for constipation or hemorrhoids,

and over 190,000 ED visits for a primary diagnosis of hemorrhoids in 2018.

Population statistics on chronic diarrhea and constipation have been the subject of global studies, and in U.S. population surveys, the prevalence rates were 5.0% (95% confidence interval [CI], 4.1%–6.0%) for chronic diarrhea and 8.8% (95% CI, 7.5%–10.0%) for chronic constipation, with women affected 1.5 times as often as men. Diarrhea and constipation are among the most common patient complaints presenting in primary care and account for nearly 50% of referrals to gastroenterologists.

Although diarrhea and constipation may present as mere nuisance symptoms at one extreme, they can be severe or life threatening at the other. Even mild symptoms may signal a serious underlying gastrointestinal (GI) lesion, such as colorectal cancer, or systemic disorder, such as thyroid disease. Given the heterogeneous causes and potential severity of these common complaints, it is imperative for clinicians to appreciate the pathophysiology, etiologic classification, diagnostic strategies, and principles of management of diarrhea and constipation so that rational and cost-effective care can be delivered.

## NORMAL PHYSIOLOGY

While the primary function of the small intestine is the digestion and assimilation of nutrients from food, the small intestine and colon together perform important functions that regulate the secretion and absorption of water and electrolytes, the storage and subsequent transport of intraluminal contents aborally, and the salvage of some nutrients that are not absorbed in the small intestine after bacterial metabolism of carbohydrate allows salvage of short-chain fatty acids. The main motor functions are summarized in **Table 49-1**. Alterations in fluid and electrolyte handling contribute significantly to diarrhea. Alterations in motor and sensory functions of the colon result in highly prevalent syndromes such as IBS, chronic diarrhea, and chronic constipation.

### NEURAL CONTROL

The small intestine and colon have intrinsic and extrinsic innervation. The *intrinsic innervation*, also called the enteric nervous system, comprises myenteric, submucosal, and mucosal neuronal layers. The function of these layers is modulated by interneurons through the actions of neurotransmitter amines or peptides, including acetylcholine, vasoactive intestinal peptide (VIP), opioids, norepinephrine, serotonin, adenosine triphosphate (ATP), and nitric oxide (NO). The myenteric plexus regulates smooth-muscle function through intermediary pacemaker-like cells called the interstitial cells of Cajal, and the submucosal plexus affects secretion, absorption, and mucosal blood flow. The enteric nervous system receives input from the extrinsic nerves, but it is capable of independent control of these functions and peristalsis.

The *extrinsic innervations* of the small intestine and colon are part of the autonomic nervous system and also modulate motor and secretory functions. The parasympathetic nerves convey visceral sensory pathways from and excitatory pathways to the small intestine and colon. Parasympathetic fibers via the vagus nerve reach the small intestine and proximal colon along the branches of the superior mesenteric artery. The distal colon is supplied by sacral parasympathetic nerves ($S_{2-4}$) via the pelvic plexus; these fibers course through the wall of the colon as ascending intracolonic fibers as far as the proximal colon. The chief excitatory neurotransmitters controlling motor function are acetylcholine and the tachykinins, such as substance P. The sympathetic nerves reach the small intestine and colon along their arterial vessels. Sympathetic input to the gut is generally excitatory to sphincters and inhibitory to nonsphincteric muscle. Visceral afferents convey sensation from the gut to the central nervous system (CNS). Some afferent fibers synapse in the prevertebral ganglia and reflexly modulate intestinal motility, blood flow, and secretion.

**TABLE 49-1 Normal Gastrointestinal Motility: Functions at Different Anatomic Levels**

| |
|---|
| **Stomach and Small Bowel** |
| Synchronized MMC in fasting |
| Accommodation, trituration, mixing, transit |
| Stomach ~3 h |
| Small bowel ~3 h |
| Ileal reservoir empties boluses |
| **Colon: Irregular Mixing, Fermentation, Absorption, Transit** |
| Cecum, ascending, transverse: reservoirs |
| Descending: conduit |
| Sigmoid/rectum: volitional reservoir |

*Abbreviation:* MMC, migrating motor complex.

### INTESTINAL FLUID ABSORPTION AND SECRETION

On an average day, 9 L of fluid enter the GI tract, ~1 L of residual fluid reaches the colon, and the stool excretion of fluid constitutes about 0.2 L/d. The colon has a large capacitance and functional reserve and may recover up to four times the usual recovered volume of 0.8 L/d, provided the rate of flow permits reabsorption to occur. Thus, the colon can partially compensate for excess fluid delivery to the colon that may result from intestinal absorptive or secretory disorders.

In the small intestine and colon, sodium absorption is predominantly electrogenic (i.e., it can be measured as an ionic current across the membrane because there is not an equivalent loss of a cation from the cell), and uptake takes place at the apical membrane; it is compensated for by the export functions of the basolateral sodium pump. Several active transport proteins are at the apical membrane, especially in the small intestine, whereby sodium ion entry is coupled to monosaccharides (e.g., glucose through the transporter SGLT1 or fructose through GLUT-5). Glucose then exits the basal membrane through a specific transport protein, GLUT-2, creating a glucose concentration and osmotic gradient between the lumen and the intercellular space, drawing water and electrolytes passively from the lumen. Several channels mediate the secretion of chloride ions in diarrheal diseases or in response to medications administered for the treatment of constipation. The diverse ion channels (chloride channels and cystic fibrosis transmembrane regulator), transporters (SGLT1, GLUT-2), and receptors (e.g., guanylate cyclase C receptor) are summarized in **Fig. 49-1**.

A variety of neural and nonneural mediators regulate colonic fluid and electrolyte balance, including cholinergic, adrenergic, and serotonergic mediators. Angiotensin and aldosterone also influence colonic absorption, reflecting the common embryologic development of the distal colonic epithelium and the renal tubules.

### SMALL-INTESTINAL MOTILITY

During the fasting period, the motility of the small intestine is characterized by a cyclical event called the migrating motor complex (MMC), which serves to clear nondigestible residue from the small intestine (the intestinal "housekeeper"). This organized, propagated series of contractions lasts, on average, 4 min, occurs every 60–90 min, and usually involves the entire small intestine. After food ingestion, the small intestine produces irregular, mixing contractions of relatively low amplitude, except in the distal ileum where more powerful contractions occur intermittently and empty the ileum by bolus transfers.

### ILEOCOLONIC STORAGE AND SALVAGE

The distal ileum acts as a reservoir, emptying intermittently by bolus movements. This action allows time for salvage of fluids, electrolytes, and nutrients. Segmentation by haustra compartmentalizes the colon and facilitates mixing, retention of residue, and formation of solid stools. There is increased appreciation of the intimate interaction between the colonic function and the luminal ecology. The resident microorganisms, predominantly anaerobic bacteria, in the colon are necessary for the digestion of unabsorbed carbohydrates that reach the colon even in health, thereby providing a vital source of nutrients to the mucosa. Normal intestinal flora also keeps pathogens at bay by

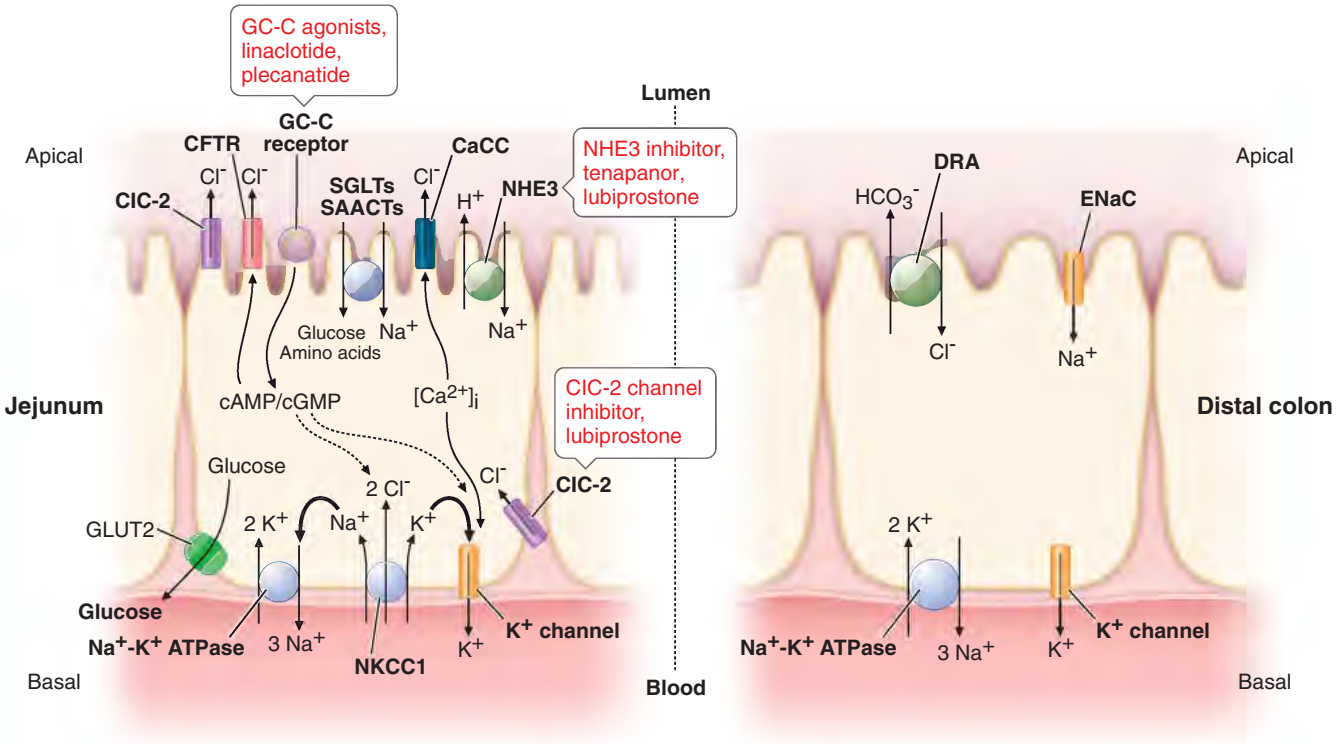


**FIGURE 49-1 Important ion transport mechanisms in the jejunum and colon, and the site of action of medications used as secretagogues in the treatment of chronic constipation.** CFTR, cystic fibrosis transmembrane regulator; CIC2, type 2 chloride channel, DRA, downregulated in adenoma (also called SLC26A3); ENaC, epithelial sodium channel; GC-C, guanylate cyclase C; $Na^+$-$K^+$ ATPase, sodium-potassium adenosine triphosphatase; NHE3, sodium-hydrogen exchanger; NKCC1, Na-K-Cl cotransporter; SAACT, sodium amino acid co-transporters; SGLT, sodium glucose transporters.

a variety of mechanisms including a crucial role in the development and maintenance of a potent but well-regulated immune response capacity to pathogens and tolerance to normal ingesta. In health, the ascending and transverse regions of colon function as reservoirs (average transit time, 15 h), and the descending colon acts as a conduit (average transit time, 3 h). The colon is efficient at conserving sodium and water, a function that is particularly important in sodium-depleted patients in whom the small intestine alone is unable to maintain sodium balance. Diarrhea or constipation may result from alteration in the reservoir function of the proximal colon or the propulsive function of the left colon. Constipation may also result from disturbances of the rectal or sigmoid reservoir, typically as a result of dysfunction of the pelvic floor, the anal sphincters, and the coordination of defecation.

## COLONIC MOTILITY AND TONE

The small-intestinal MMC only rarely continues into the colon. However, short duration or phasic antegrade and retrograde contractions mix colonic contents, and high-amplitude (>75 mmHg) propagated contractions (HAPCs) are sometimes associated with mass movements through the colon and normally occur approximately five times per day, usually on awakening in the morning and postprandially. Increased frequency of HAPCs may result in diarrhea or urgency.

Colonic tone refers to the background contractility upon which phasic contractile activity is superimposed. It is an important cofactor in the colon's capacitance (volume accommodation) and sensation.

## COLONIC MOTILITY AFTER MEAL INGESTION

After meal ingestion, colonic phasic and tonic contractility increase for a period of ~2 h. The initial phase (~10 min) is mediated by the vagus nerve in response to mechanical distention of the stomach. The subsequent response of the colon requires caloric stimulation (e.g., intake of at least 500 kcal) and is mediated, at least in part, by hormones (e.g., motilin and cholecystokinin).

## DEFECATION

Tonic contraction of the puborectalis muscle, which forms a sling around the rectoanal junction, is important to maintain continence; during defecation, sacral parasympathetic nerves relax this muscle, facilitating the straightening of the rectoanal angle (Fig. 49-2).

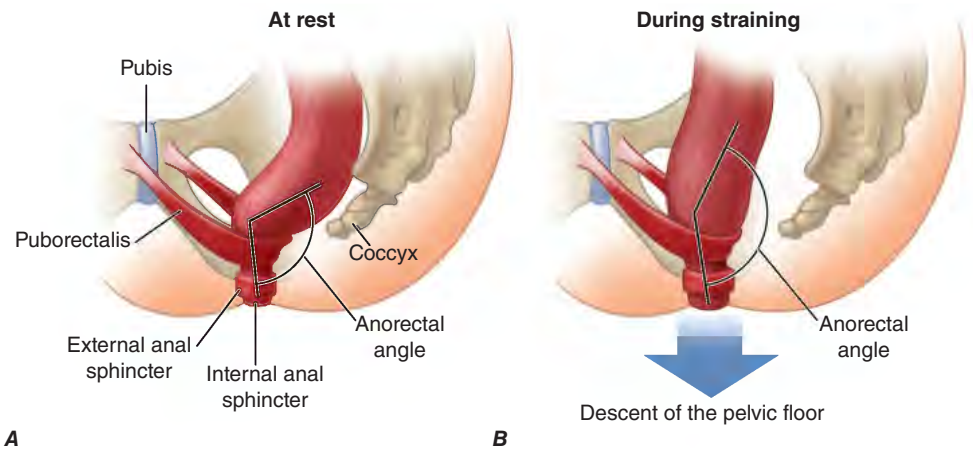


**FIGURE 49-2 Sagittal view of the anorectum (*A*) at rest and (*B*) during straining to defecate.** Continence is maintained by normal rectal sensation and tonic contraction of the internal anal sphincter and the puborectalis muscle, which wraps around the anorectum, maintaining an anorectal angle between 80° and 110°. During defecation, the pelvic floor muscles (including the puborectalis) relax, allowing the anorectal angle to straighten by at least 15°, and the perineum descends by 1–3.5 cm. The external anal sphincter also relaxes and reduces pressure on the anal canal. *(From The New England Journal of Medicine, A Lembo, M Camilleri; Chronic constipation, 349:1360, Copyright © 2003, Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.)*

Distention of the rectum results in transient relaxation of the internal anal sphincter via intrinsic and reflex sympathetic innervation. As sigmoid and rectum contract, and straining (Valsalva maneuver) increases intraabdominal pressure and pressure within the rectum, the rectosigmoid angle opens by >15°. Voluntary relaxation of the external anal sphincter (striated muscle innervated by the pudendal nerve) in response to the sensation produced by distention permits the evacuation of feces. Defecation can also be delayed voluntarily by contraction of the external anal sphincter.

## DIARRHEA

### DEFINITION

Diarrhea is loosely defined as passage of abnormally liquid or unformed stools at an increased frequency. For adults on a typical Western diet, stool weight >200 g/d can generally be considered diarrheal. Diarrhea may be further defined as *acute* if <2 weeks, *persistent* if 2–4 weeks, and *chronic* if >4 weeks in duration.

Two common conditions, usually associated with the passage of stool totaling <200 g/d, must be distinguished from diarrhea, because diagnostic and therapeutic algorithms differ. *Pseudodiarrhea*, or the frequent passage of small volumes of stool, is often associated with rectal urgency, tenesmus, or a feeling of incomplete evacuation and accompanies IBS or proctitis. *Fecal incontinence* is the involuntary discharge of rectal contents and is most often caused by neuromuscular disorders or structural anorectal problems. Diarrhea and urgency, especially if severe, may aggravate or cause incontinence. Pseudodiarrhea and fecal incontinence occur at prevalence rates comparable to or higher than that of chronic diarrhea and should always be considered in patients complaining of "diarrhea." Overflow diarrhea may occur in nursing home patients due to fecal impaction that is readily detectable by rectal examination. A careful history and physical examination generally allow these conditions to be discriminated from true diarrhea.

### ACUTE DIARRHEA

More than 90% of cases of acute diarrhea are caused by infectious agents; these cases are often accompanied by vomiting, fever, and abdominal pain. The remaining 10% or so are caused by medications, toxic ingestions, ischemia, food indiscretions, and other conditions.

**Infectious Agents** Domestically, infections predominate, whereas bacterial infections may be more important in travel-acquired disease. Most infectious diarrheas are acquired by fecal-oral transmission or, more commonly, via ingestion of food or water contaminated with pathogens from human or animal feces. In the immunocompetent person, the resident fecal microflora, containing >500 taxonomically distinct species, are rarely the source of diarrhea and may actually play a role in suppressing the growth of ingested pathogens. Disturbances of flora by antibiotics can lead to diarrhea by reducing the digestive function or by allowing the overgrowth of pathogens, such as *Clostridioides difficile* **(Chap. 139)**. Acute infection or injury occurs when the ingested agent overwhelms or bypasses the host's mucosal immune and nonimmune (gastric acid, digestive enzymes, mucus secretion, peristalsis, and suppressive resident flora) defenses. Established clinical associations with specific enteropathogens may offer diagnostic clues. Diarrhea occasionally is an early symptom of infection such as SARS-CoV-2 and *Legionella*.

In the United States, five high-risk groups are recognized:

1. *Travelers*. Up to 40% of U.S. tourists to endemic regions of Latin America, Africa, and Asia develop so-called traveler's diarrhea, most commonly due to enterotoxigenic or enteroaggregative *Escherichia coli* as well as to *Campylobacter*, *Shigella*, *Aeromonas*, norovirus, *Coronavirus*, and *Salmonella*. Other causative organisms in traveler's diarrhea among campers, backpackers, and swimmers are *Giardia* and *Cyclospora*. Cruise ships may be affected by outbreaks of gastroenteritis caused by agents such as norovirus.
2. *Consumers of certain foods*. Diarrhea closely following food consumption at a picnic, banquet, or restaurant may suggest infection with *Salmonella*, *Campylobacter*, or *Shigella* from chicken; enterohemorrhagic *E. coli* (O157:H7) from undercooked hamburger; *Bacillus cereus* from fried rice or other reheated food; *Staphylococcus aureus* or *Salmonella* from mayonnaise or creams; *Salmonella* from eggs; *Listeria* from fresh or frozen uncooked foods, mushrooms, or dairy products; and *Vibrio* species, *Salmonella*, or acute hepatitis A from seafood, especially if raw. State departments of public health and the National Outbreak Reporting System issue communications regarding domestic and foreign food-related illnesses, often identified by rapid DNA typing (PulseNet), that cause epidemics in the United States (e.g., the *Listeria* epidemic of 2020 from imported enoki mushrooms). Diverse foods including meat, seafood, and fresh produce and drinking water may be contaminated with *C. difficile* and the same ribotypes that actually cause the human disease. However, it has not been conclusively demonstrated that food or water actually causes community-acquired *C. difficile* infection other than a waterborne outbreak in Finland associated with contamination of a public tap water distribution system.
3. *Immunodeficient persons*. Individuals at risk for diarrhea include those with either primary immunodeficiency (e.g., IgA deficiency, common variable hypogammaglobulinemia, chronic granulomatous disease) or secondary immunodeficiency states (e.g., AIDS, senescence, pharmacologic suppression). Common enteric pathogens often cause a more severe and protracted diarrheal illness. In persons with AIDS, opportunistic infections, such as by *Mycobacterium* species, certain viruses (cytomegalovirus, adenovirus, and herpes simplex), and protozoa (*Cryptosporidium*, *Isospora belli*, microsporidia, and *Blastocystis hominis*) may also play a role **(Chap. 208)**. In patients with AIDS, agents transmitted venereally per rectum or by extension from vaginal infection (e.g., *Neisseria gonorrhoeae*, *Treponema pallidum*, *Chlamydia*) may contribute to proctocolitis. Persons with hemochromatosis are especially prone to invasive, even fatal, enteric infections with *Vibrio* species and *Yersinia* infections and should avoid raw fish and exposing open wounds to seawater.
4. *Daycare attendees and their family members*. Infections with *Shigella*, *Giardia*, *Cryptosporidium*, rotavirus, and other agents are very common.
5. *Institutionalized persons*. Infectious diarrhea is one of the most frequent nosocomial infections in hospitals and long-term care facilities; the causes are a variety of microorganisms but most commonly *C. difficile*. *C. difficile* can affect those with no history of antibiotic use and is often community acquired.

The pathophysiology underlying acute diarrhea by infectious agents results in specific clinical features that may be helpful in diagnosis **(Table 49-2)**. Profuse, watery diarrhea secondary to small-bowel hypersecretion occurs with ingestion of preformed bacterial toxins, enterotoxin-producing bacteria, and enteroadherent pathogens. Abrupt onset of diarrhea associated with marked vomiting and minimal or no fever within a few hours after ingestion suggests a toxin etiology; vomiting is usually less, abdominal cramping or bloating is greater, and fever is higher with enteroadherent pathogens. Cytotoxin-producing and invasive microorganisms all cause high fever and abdominal pain. Invasive bacteria and *Entamoeba histolytica* often cause bloody diarrhea (referred to as *dysentery*). *Yersinia* invades the terminal ileal and proximal colon mucosa and may cause especially severe abdominal pain with tenderness mimicking acute appendicitis.

Finally, infectious diarrhea may be associated with systemic manifestations. Reactive arthritis or the combination of arthritis, urethritis, and conjunctivitis (Reiter's syndrome) may accompany or follow infections by *Salmonella*, *Campylobacter*, *Shigella*, and *Yersinia*. Yersiniosis may also lead to an autoimmune-type thyroiditis, pericarditis, and glomerulonephritis. Both enterohemorrhagic *E. coli* (O157:H7) and *Shigella* can lead to the *hemolytic-uremic syndrome* with an attendant

**TABLE 49-2 Association Between Pathobiology of Causative Agents and Clinical Features in Acute Infectious Diarrhea**

| PATHOBIOLOGY/AGENTS | INCUBATION PERIOD | VOMITING | ABDOMINAL PAIN | FEVER | DIARRHEA |
|---|---|---|---|---|---|
| Toxin producers | | | | | |
| Preformed toxin | | | | | |
| *Bacillus cereus*, *Staphylococcus aureus*, *Clostridium perfringens* | 1–8 h<br>8–24 h | 3–4+ | 1–2+ | 0–1+ | 3–4+, watery |
| Enterotoxin | | | | | |
| *Vibrio cholerae*, enterotoxigenic *Escherichia coli*, *Klebsiella pneumoniae*, *Aeromonas* species | 8–72 h | 2–4+ | 1–2+ | 0–1+ | 3–4+, watery |
| Enteroadherent | | | | | |
| Enteropathogenic and enteroadherent *E. coli*, *Giardia* organisms, cryptosporidiosis, helminths | 1–8 d | 0–1+ | 1–3+ | 0–2+ | 1–2+, watery, mushy |
| Cytotoxin producers | | | | | |
| *Clostridioides difficile* | 1–3 d | 0–1+ | 3–4+ | 1–2+ | 1–3+, usually watery, occasionally bloody |
| Hemorrhagic *E. coli* | 12–72 h | 0–1+ | 3–4+ | 1–2+ | 1–3+, initially watery, quickly bloody |
| Invasive organisms | | | | | |
| Minimal inflammation | | | | | |
| Rotavirus and norovirus | 1–3 d | 1–3+ | 2–3+ | 3–4+ | 1–3+, watery |
| Variable inflammation | | | | | |
| *Salmonella*, *Campylobacter*, and *Aeromonas* species, *Vibrio parahaemolyticus*, *Yersinia* | 12 h–11 d | 0–3+ | 2–4+ | 3–4+ | 1–4+, watery or bloody |
| Severe inflammation | | | | | |
| *Shigella* species, enteroinvasive *E. coli*, *Entamoeba histolytica* | 12 h–8 d | 0–1+ | 3–4+ | 3–4+ | 1–2+, bloody |

*Source:* Adapted from DW Powell, in T Yamada (ed): *Textbook of Gastroenterology and Hepatology*, 4th ed. Philadelphia, Lippincott Williams & Wilkins, 2003.

high mortality rate. The syndrome of postinfectious IBS has now been recognized as a complication of infectious diarrhea, particularly *Campylobacter jejuni* infection. Similarly, acute gastroenteritis may precede the diagnosis of celiac disease or Crohn's disease, but the risk of developing these diseases is low, and there is no indication to screen for these diseases (e.g., by serology or endoscopy) during the episode of acute gastroenteritis. Acute diarrhea can also be a major symptom of several systemic infections including *viral hepatitis*, *listeriosis*, *legionellosis*, and *toxic shock syndrome*.

**Other Causes** Side effects from *medications* are probably the most common noninfectious causes of acute diarrhea, and etiology may be suggested by a temporal association between use and symptom onset. Although innumerable medications may produce diarrhea, some of the more frequently incriminated include antibiotics, cardiac antidysrhythmics, antihypertensives, nonsteroidal anti-inflammatory drugs (NSAIDs), certain antidepressants, weight loss agents, chemotherapeutic agents, bronchodilators, antacids, and laxatives.

Occlusive or nonocclusive *ischemic colitis* typically occurs in persons aged >50 years; often presents as acute lower abdominal pain preceding watery, then bloody diarrhea; and generally results in acute inflammatory changes in the sigmoid or left colon while sparing the rectum. Acute diarrhea may accompany colonic diverticulitis and graft-versus-host disease.

Acute diarrhea, often associated with systemic compromise, can follow *ingestion of toxins* including organophosphate insecticides, amanita and other mushrooms, arsenic, and preformed toxins in seafood such as ciguatera (from algae that the fish eat) and scombroid (an excess of histamine due to inadequate refrigeration). Acute allergic reaction to food ingestion, including galactose alpha-1,3-galactose (alpha gal) contained in meat, can have a similar presentation including systemic collapse, which should not be mistaken for simple dehydration secondary to diarrhea. *Conditions causing chronic diarrhea* can also be confused with acute diarrhea early in their course. This confusion may occur with inflammatory bowel disease (IBD) and some of the other inflammatory chronic diarrheas that may have an abrupt rather than insidious onset and exhibit features that mimic infection.

## APPROACH TO THE PATIENT

### Acute Diarrhea

The decision to evaluate acute diarrhea depends on its severity and duration and on various host factors **(Fig. 49-3)**. Most episodes of acute diarrhea are mild and self-limited and do not justify the cost and potential morbidity rate of diagnostic or pharmacologic interventions. Indications for evaluation (stool microbiological studies) include profuse diarrhea with dehydration, overtly bloody stools, fever ≥38.5°C (≥101°F), duration >48 h without improvement, recent antibiotic use, new community outbreaks, associated severe abdominal pain in patients aged >50 years, and elderly (≥70 years) or immunocompromised patients. In some cases of moderately severe febrile diarrhea associated with fecal leukocytes (or increased fecal levels of the leukocyte proteins, such as calprotectin) or with overt blood, a diagnostic evaluation might be avoided in favor of an empirical antibiotic trial (see below).

The cornerstone of diagnosis in those suspected of severe acute infectious diarrhea acquired domestically is microbiologic analysis of the stool. Workup for a microbiologic cause now starts with culture-independent methods. These methods, relying on the identification of unique DNA sequences, are more rapid, sensitive, specific, and cost-effective. The less sensitive culture-based methods for bacteria, direct inspection for ova and parasites, and immunoassays for certain bacterial toxins (*C. difficile*), viral antigens (rotavirus), and protozoal antigens (*Giardia*, *E. histolytica*) are now primarily used when multiplexed tests are not available or for outbreak detection that still relies on isolate cultures and should be done when an outbreak is suspected or in domestically acquired severe diarrhea. Alternatively, the aforementioned clinical and epidemiologic associations may assist in focusing the evaluation. If a particular pathogen or set of possible pathogens is so implicated, either the whole panel of routine studies may not be necessary or, in some instances, special cultures may be appropriate, as for enterohemorrhagic and other types of *E. coli*, *Vibrio* species, and *Yersinia*. There is no indication for serologic or endoscopic evaluation to exclude underlying intestinal diseases in the context of acute diarrhea.

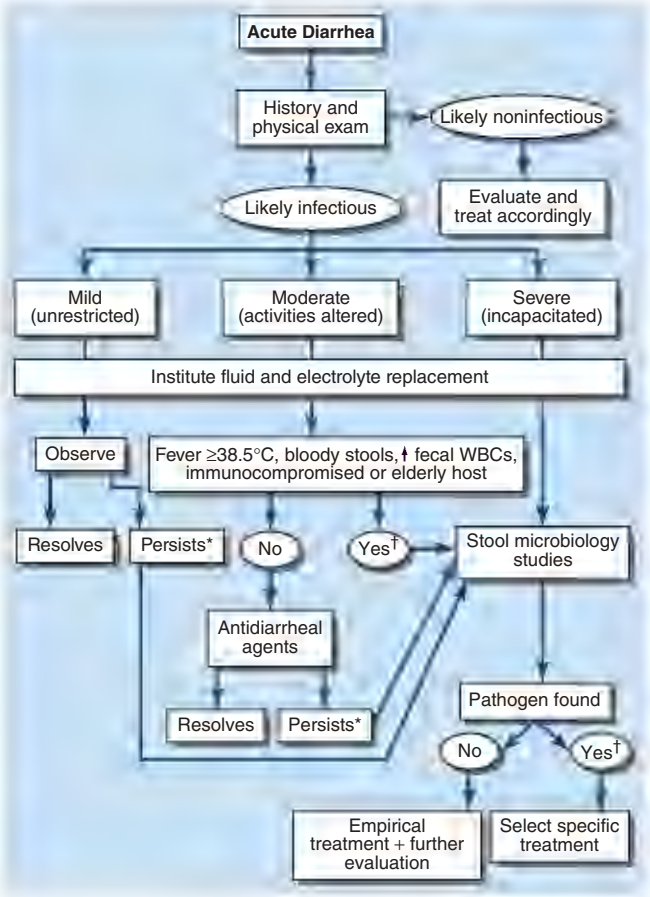


**FIGURE 49-3 Algorithm for the management of acute diarrhea.** Consider empirical treatment before evaluation with (*) metronidazole and with (†) quinolone. WBCs, white blood cells.

Persistent diarrhea is commonly due to *Giardia* **(Chap. 230)**, but additional causative organisms that should be considered include *C. difficile* (especially if antibiotics had been administered), *E. histolytica*, *Cryptosporidium*, *Campylobacter*, and others. If stool studies are unrevealing and if the diarrhea continues to become chronic (>30 days), flexible sigmoidoscopy with biopsies and upper endoscopy with duodenal aspirates and biopsies may be indicated to exclude celiac or Crohn's disease, but these should not be performed during or early after acute enteritis because of risk of false-positive results. Brainerd diarrhea is an increasingly recognized entity characterized by an abrupt-onset diarrhea that persists for at least 4 weeks, but may last 1–3 years, and is thought to be of infectious origin. It may be associated with subtle inflammation of the distal small intestine or proximal colon.

Structural examination by sigmoidoscopy, colonoscopy, or abdominal computed tomography (CT) scanning (or other imaging approaches) should only be used for patients with uncharacterized persistent diarrhea when other symptoms suggest IBD or as an initial approach in patients with suspected noninfectious acute diarrhea such as might be caused by ischemic colitis, diverticulitis, or partial bowel obstruction. However, acute diarrhea may be associated with finding of air-fluid levels on abdominal radiograph, necessitating CT imaging to exclude organic obstruction.

## TREATMENT

### Acute Diarrhea

Fluid and electrolyte replacement are of central importance to all forms of acute diarrhea. Fluid replacement alone may suffice for mild cases. Oral sugar-electrolyte solutions (iso-osmolar sport drinks or designed formulations) should be instituted promptly with severe diarrhea to limit dehydration or in elderly or vulnerable patients, which is the major cause of death. Profoundly dehydrated patients, especially infants and the elderly, require IV rehydration.

In moderately severe nonfebrile and nonbloody diarrhea, antimotility and antisecretory agents such as loperamide can be useful adjuncts to control symptoms. Such agents should be avoided with febrile dysentery, which may be prolonged by them, and should be used with caution with drugs that increase levels due to cardiac arrhythmia (due to effect on cardiac transmembrane ion channels). Bismuth subsalicylate may reduce symptoms of vomiting and diarrhea but should not be used to treat immunocompromised patients or those with renal impairment because of the risk of bismuth encephalopathy.

Judicious use of antibiotics is appropriate in selected instances of acute diarrhea, especially in traveler's diarrhea, and may reduce its severity and duration (Fig. 49-3). Many physicians treat moderately to severely ill patients with febrile diarrheal illness empirically without diagnostic evaluation using a macrolide such as azithromycin (single dose of 500 mg) or a quinolone such as ciprofloxacin (500 mg bid for 3–5 d). Empirical treatment can also be considered for suspected giardiasis with metronidazole (250 mg qid for 7 d). Selection of antibiotics and dosage regimens are otherwise dictated by specific pathogens, geographic patterns of resistance, and conditions found **(Chaps. 138, 166, and 170–176)**. Because of resistance to first-line treatments, newer agents such as nitazoxanide (500 mg bid for 3 d) may be required for *Giardia* and *Cryptosporidium* infections. Antibiotic coverage is indicated, whether or not a causative organism is discovered, in patients who are immunocompromised, have mechanical heart valves or recent vascular grafts, or are elderly. Bismuth subsalicylate may reduce the frequency of traveler's diarrhea. Antibiotic prophylaxis is only indicated for certain patients traveling to high-risk countries in whom the likelihood or seriousness of acquired diarrhea would be especially high, including those with immunocompromise, IBD, hemochromatosis, or gastric achlorhydria. Use of ciprofloxacin, azithromycin, or rifaximin may reduce bacterial diarrhea by 90% in such travelers. Rifaximin is not suitable for invasive disease but rather as treatment for uncomplicated traveler's diarrhea. There is little role for endoscopic evaluation in most circumstances except in immunocompromised patients. Finally, physicians should be vigilant to identify if an outbreak of diarrheal illness and to alert the public health authorities promptly in order to reduce further spread.

## CHRONIC DIARRHEA

Diarrhea lasting >4 weeks warrants evaluation to exclude serious underlying pathology. In contrast to acute diarrhea, most of the causes of chronic diarrhea are noninfectious. The classification of chronic diarrhea by pathophysiologic mechanism facilitates a rational approach to management, although many diseases cause diarrhea by more than one mechanism **(Table 49-3)**.

**Secretory Causes** Secretory diarrheas are due to derangements in fluid and electrolyte transport across the enterocolonic mucosa. They are characterized by watery, large-volume fecal outputs that are typically painless and persist with fasting. Because there is no malabsorbed solute, stool osmolality shows no fecal osmotic gap.

**MEDICATIONS** Side effects from regular ingestion of drugs and toxins are the most common secretory causes of chronic diarrhea. Hundreds of prescription and over-the-counter medications (see earlier section, "Acute Diarrhea, Other Causes") may produce diarrhea. Surreptitious or habitual use of stimulant laxatives (e.g., senna, cascara, bisacodyl, ricinoleic acid [castor oil]) must also be considered. Chronic ethanol consumption may cause a secretory-type diarrhea due to enterocyte injury with impaired sodium and water absorption, rapid transit, and other alterations. Inadvertent ingestion of certain environmental toxins (e.g., arsenic) may lead to chronic rather than acute diarrhea. Certain bacterial infections may occasionally persist and be associated with a secretory-type diarrhea. The oral angiotensin receptor blockers, including olmesartan, are associated with diarrhea due to sprue-like

**TABLE 49-3 Major Causes of Chronic Diarrhea According to Predominant Pathophysiologic Mechanism**

| |
|---|
| **Secretory Causes** |
| Exogenous stimulant laxatives |
| Chronic ethanol ingestion |
| Other drugs and toxins |
| Endogenous laxatives (dihydroxy bile acids) |
| Idiopathic secretory diarrhea or bile acid diarrhea |
| Certain bacterial infections |
| Bowel resection, disease, or fistula (↓ absorption) |
| Partial bowel obstruction or fecal impaction |
| Hormone-producing tumors (carcinoid, VIPoma, medullary cancer of thyroid, mastocytosis, gastrinoma, colorectal villous adenoma) |
| Addison's disease |
| Congenital electrolyte absorption defects |
| **Osmotic Causes** |
| Osmotic laxatives ($Mg^{2+}$, $PO_4^{-3}$, $SO_4^{-2}$) |
| Lactase and other disaccharide deficiencies |
| Nonabsorbable carbohydrates (fructose, sorbitol, lactulose, polyethylene glycol) |
| Wheat and FODMAP intolerance |
| **Steatorrheal Causes** |
| Intraluminal maldigestion (pancreatic exocrine insufficiency, bacterial overgrowth, bariatric surgery, liver disease) |
| Mucosal malabsorption (celiac disease, Whipple's disease, infections, abetalipoproteinemia, ischemia, drug-induced enteropathy) |
| Postmucosal obstruction (1° or 2° lymphatic obstruction), amyloidosis |
| **Inflammatory Causes** |
| Idiopathic inflammatory bowel disease (Crohn's, chronic ulcerative colitis) |
| Lymphocytic and collagenous colitis |
| Immune-related mucosal disease (1° or 2° immunodeficiencies, food allergy, eosinophilic gastroenteritis, C1-esterase inhibitor deficiency, graft-versus-host disease) |
| Infections (invasive bacteria, viruses, and parasites, Brainerd diarrhea) |
| Radiation injury |
| Gastrointestinal malignancies |
| **Dysmotile Causes** |
| Irritable bowel syndrome (including postinfectious IBS) |
| Visceral neuromyopathies |
| Hyperthyroidism |
| Drugs (prokinetic agents) and poisons |
| Postvagotomy |
| **Factitial Causes** |
| Munchausen |
| Eating disorders |
| **Iatrogenic Causes** |
| Cholecystectomy |
| Ileal resection |
| Bariatric surgery |
| Vagotomy, fundoplication |

*Abbreviations:* FODMAP, fermentable oligosaccharides, disaccharides, monosaccharides, and polyols; IBS, irritable bowel syndrome.

enteropathy, and more rarely, angiotensin-converting enzyme inhibitors are associated with intestinal angioedema. Diarrhea is also a relatively infrequent adverse effect of treatment with glucagon-like peptide 1 receptor agonists (e.g., semaglutide), although the mechanism in humans has not been elucidated.

**BOWEL RESECTION OR BYPASS, MUCOSAL DISEASE, OR ENTEROCOLIC FISTULA** These conditions may result in a secretory-type diarrhea because of inadequate surface for reabsorption of secreted fluids and electrolytes. Unlike other secretory diarrheas, this subset of conditions tends to worsen with eating. With disease (e.g., Crohn's ileitis) or resection of <100 cm of terminal ileum, dihydroxy bile acids may escape absorption and stimulate colonic secretion (cholerheic diarrhea). This mechanism may contribute to so-called *idiopathic secretory diarrhea or bile acid diarrhea (BAD)*, in which bile acids are functionally malabsorbed from a normal-appearing terminal ileum. This may account for an average of 40% of unexplained chronic diarrhea and about 1% of the general population (same as celiac disease). Reduced negative feedback regulation of bile acid synthesis in hepatocytes by fibroblast growth factor 19 (FGF-19) produced by ileal enterocytes results in a degree of bile-acid synthesis that exceeds the normal capacity for ileal reabsorption, producing BAD. BAD is characterized by severe diarrhea, urgency, and impaired quality of life in part due to the need to be close to toilet facilities. It can be positively diagnosed by elevated fasting serum 7αC4 or fecal primary or total bile acids.

Partial bowel obstruction, ostomy stricture, or fecal impaction may paradoxically lead to increased fecal output due to fluid hypersecretion.

**HORMONES** Although uncommon, secretory diarrhea may be mediated by hormones. *Metastatic gastrointestinal carcinoid tumors* or, rarely, *primary bronchial carcinoids* may produce watery diarrhea alone or as part of the carcinoid syndrome that comprises episodic flushing, wheezing, dyspnea, and right-sided valvular heart disease. Diarrhea is due to the release into the circulation of potent intestinal secretagogues including serotonin, histamine, prostaglandins, and various kinins. Pellagra-like skin lesions may rarely occur as the result of serotonin overproduction with niacin depletion. *Gastrinoma* typically presents with refractory peptic ulcers, but diarrhea occurs in up to one-third of cases and may be the only clinical manifestation in 10%. Diarrhea most often results from fat maldigestion owing to pancreatic enzyme inactivation by low intraduodenal pH secondary to gastric acid hypersecretion. The watery diarrhea, hypokalemia, achlorhydria syndrome, also called *pancreatic cholera*, is due to secretion of VIP and a host of other peptide hormones by a non-β cell pancreatic adenoma, referred to as a *VIPoma*. The secretory diarrhea is often massive, with stool volumes >3 L/d; daily volumes as high as 20 L have been reported. Life-threatening dehydration; neuromuscular dysfunction from associated hypokalemia, hypomagnesemia, or hypercalcemia; flushing; and hyperglycemia may accompany a VIPoma. *Medullary carcinoma of the thyroid* may present with watery diarrhea caused by calcitonin, other secretory peptides, or prostaglandins. Prominent diarrhea is often associated with metastatic disease and poor prognosis. *Systemic mastocytosis*, which may be associated with the skin lesion urticaria pigmentosa, may cause diarrhea that is either secretory and mediated by histamine or inflammatory due to intestinal infiltration by mast cells. Large *colorectal villous adenomas* may rarely be associated with a secretory diarrhea that may cause hypokalemia; these effects may be mediated by prostaglandins and can be inhibited by NSAIDs.

**CONGENITAL DEFECTS IN ION ABSORPTION** Rarely, defects in specific carriers associated with ion absorption cause watery diarrhea from birth. These disorders include defective $Cl^-/HCO_3^-$ exchange (*congenital chloridorrhea*) with alkalosis (which results from a mutated *DRA* [down-regulated in adenoma] gene) and defective $Na^+/H^+$ exchange (*congenital sodium diarrhea*), which results from a mutation in the *NHE3* (sodium-hydrogen exchanger) gene and results in acidosis.

Some hormone deficiencies may be associated with watery diarrhea, such as occurs with adrenocortical insufficiency (Addison's disease).

**Osmotic Causes** Osmotic diarrhea occurs when ingested, poorly absorbable, osmotically active solutes draw enough fluid into the lumen to exceed the reabsorptive capacity of the colon. Fecal water output increases in proportion to such a solute load. Osmotic diarrhea characteristically ceases with fasting or with discontinuation of the causative agent.

**OSMOTIC LAXATIVES** Ingestion of magnesium-containing antacids, health supplements, or laxatives may induce osmotic diarrhea typified by a stool osmotic gap (>50 mosmol/L), which is calculated by

serum osmolarity (typically 290 mosmol/kg) − (2 × [fecal sodium + potassium concentration]). Measurement of fecal osmolarity may be erroneous because carbohydrates are metabolized by colonic bacteria, causing an increase in osmolarity.

**CARBOHYDRATE MALABSORPTION** Carbohydrate malabsorption due to acquired or congenital defects in brush-border disaccharidases and other enzymes leads to osmotic diarrhea with a low fecal pH. One of the most common causes of chronic diarrhea in adults is *lactase deficiency*, which affects three-fourths of nonwhites worldwide and 5–30% of persons in the United States; the total lactose load at any one time influences the symptoms experienced. Most patients learn to avoid milk products without requiring treatment with enzyme supplements. Sugars such as sorbitol, lactulose, or fructose ingested with medications, gum, or high fructose–containing beverages or candies may be incompletely absorbed, and diarrhea ensues.

**FODMAP AND WHEAT INTOLERANCE** Chronic diarrhea, bloating, and abdominal pain are recognized as symptoms of nonceliac wheat intolerance (which is associated with impaired intestinal or colonic barrier function) and intolerance of fermentable oligosaccharides, disaccharides, monosaccharides, and polyols (FODMAPs). The latter's effects represent the interaction between the GI microbiome and the nutrients.

**Steatorrheal Causes** Fat malabsorption may lead to greasy, foul-smelling, difficult-to-flush diarrhea often associated with weight loss and nutritional deficiencies due to concomitant malabsorption of amino acids and vitamins. Increased fecal output is caused by the osmotic effects of fatty acids, especially after bacterial hydroxylation, and, to a lesser extent, by the neutral fat. Whereas the normal fecal fat is <7 g/d on a 100 g/d fat diet, rapid-transit diarrhea may result in fecal fat up to 14 g/d of fat; daily fecal fat averages 15–25 g with small-intestinal diseases and is often >32 g with pancreatic exocrine insufficiency. Intraluminal maldigestion, mucosal malabsorption, or lymphatic obstruction may produce steatorrhea.

**INTRALUMINAL MALDIGESTION** This condition most commonly results from pancreatic exocrine insufficiency, which occurs when >90% of pancreatic secretory function is lost. *Chronic pancreatitis*, usually a sequel of ethanol abuse, most frequently causes pancreatic insufficiency. Other causes include medications, *cystic fibrosis*, *pancreatic duct obstruction*, and, rarely, *somatostatinoma*. Bacterial overgrowth in the small intestine may deconjugate bile acids and alter micelle formation, impairing fat digestion; it occurs with stasis from a blind-loop, multiple small-bowel diverticula, or dysmotility with intestinal dilatation. Finally, cirrhosis or biliary obstruction may lead to mild steatorrhea due to deficient intraluminal bile acid concentration.

**MUCOSAL MALABSORPTION** Mucosal malabsorption occurs from a variety of enteropathies, most commonly *celiac disease*. This gluten-sensitive enteropathy affects all ages and is characterized by villous atrophy and crypt hyperplasia in the proximal small bowel and can present with steatorrhea with multiple nutritional deficiencies of varying severity. Celiac disease, which is more frequent than previously thought, affects ~1% of the population, frequently presents without steatorrhea, can mimic IBS, and has many other GI and extraintestinal manifestations. *Tropical sprue* may produce a similar histologic and clinical syndrome but occurs in residents of or travelers to tropical climates; abrupt onset and response to antibiotics suggest an infectious etiology. *Whipple's disease*, due to the bacillus *Tropheryma whipplei* and macrophage infiltration of the small-bowel mucosa, is a far less common cause of steatorrhea that typically occurs in young or middle-aged men; it is frequently associated with arthralgias, fever, lymphadenopathy, and extreme fatigue, and it may affect the CNS and endocardium. A similar clinical and histologic picture results from *Mycobacterium avium-intracellulare* infection in patients with AIDS. *Abetalipoproteinemia* is a rare defect of chylomicron formation and fat malabsorption in children. It is associated with acanthocytic erythrocytes, ataxia, and retinitis pigmentosa. Several other conditions may cause mucosal malabsorption including infections, especially with protozoa such as *Giardia*, numerous medications (e.g., olmesartan, mycophenolate mofetil, colchicine, cholestyramine, neomycin), idiopathic enteropathies, lymphoproliferative disorders, amyloidosis, and chronic ischemia.

**POSTMUCOSAL LYMPHATIC OBSTRUCTION** This may result from rare *congenital intestinal lymphangiectasia* or from *acquired lymphatic obstruction* secondary to trauma, tumor, cardiac disease, or infection, and leads to the unique fat malabsorption with enteric losses of protein (often causing edema) and lymphocytopenia. Carbohydrate and amino acid absorption are preserved.

**Inflammatory Causes** Inflammatory diarrheas are generally accompanied by pain, fever, bleeding, or other manifestations of inflammation. The mechanism of diarrhea may not only be exudation but, depending on lesion site, may include fat malabsorption, disrupted fluid/electrolyte absorption, and hypersecretion or hypermotility from release of cytokines and other inflammatory mediators. Stool analysis reveals leukocytes or leukocyte-derived proteins such as calprotectin. With severe inflammation, exudative protein loss can lead to anasarca (generalized edema). Any middle-aged or older person with chronic inflammatory-type diarrhea, especially with blood, should be carefully evaluated (e.g., with colonoscopy) to exclude a colorectal tumor.

**IDIOPATHIC INFLAMMATORY BOWEL DISEASE** *Crohn's disease* and *chronic ulcerative colitis* are among the most common organic causes of chronic diarrhea in adults and range in severity from mild to fulminant and life-threatening. They may be associated with uveitis, polyarthralgias, oral ulceration, cholestatic liver disease (primary sclerosing cholangitis), and skin lesions (erythema nodosum, pyoderma gangrenosum). *Microscopic colitis*, including both lymphocytic and *collagenous colitis*, may cause chronic watery diarrhea, especially in middle-aged women and those on NSAIDs, statins, proton pump inhibitors (PPIs), and selective serotonin reuptake inhibitors (SSRIs); biopsy of an otherwise normal-appearing colon is required for histologic diagnosis. It may coexist with symptoms suggesting IBS or with celiac sprue or drug-induced enteropathy. It may be associated with BAD. It typically responds well to anti-inflammatory drugs (e.g., bismuth), the opioid agonist loperamide, or budesonide or bile acid sequestrants.

**PRIMARY OR SECONDARY FORMS OF IMMUNODEFICIENCY** Immunodeficiency may lead to prolonged infectious diarrhea. With selective IgA deficiency or common variable *hypogammaglobulinemia*, diarrhea is particularly prevalent and often the result of giardiasis, bacterial overgrowth, or sprue or *Strongyloides stercoralis* acquired in the tropics or subtropics.

**EOSINOPHILIC GASTROENTERITIS** Eosinophil infiltration of the mucosa, muscularis, or serosa at any level of the GI tract may cause diarrhea, pain, vomiting, or ascites. Affected patients often have an atopic history, Charcot-Leyden crystals due to extruded eosinophil contents may be seen on microscopic inspection of stool, and peripheral eosinophilia is present in 50–75% of patients. While hypersensitivity to certain foods occurs in adults, true food allergy causing chronic diarrhea is rare.

**OTHER CAUSES** Chronic inflammatory diarrhea may be caused by *radiation enterocolitis*, *chronic graft-versus-host disease*, autoimmune or idiopathic enteropathies, *Behçet's syndrome*, and *Cronkhite-Canada syndrome*, among others.

**Dysmotility Causes** Rapid transit may accompany many diarrheas as a secondary or contributing phenomenon, but primary dysmotility is an unusual etiology of true diarrhea. Stool features often suggest a secretory diarrhea, but mild steatorrhea of up to 14 g of fat per day can be produced by maldigestion from rapid transit alone. *Hyperthyroidism*, *carcinoid syndrome*, and certain drugs (e.g., prostaglandins, prokinetic agents, cholinesterase inhibitors) may produce hypermotility with resultant diarrhea. Primary visceral neuromyopathies or idiopathic acquired intestinal pseudoobstruction may lead to stasis with secondary bacterial overgrowth causing diarrhea. *Diabetic diarrhea*, often accompanied by peripheral and generalized autonomic neuropathies, may occur in part because of intestinal dysmotility.

The exceedingly common IBS (10% point prevalence, 1–2% per year incidence) is characterized by disturbed intestinal and colonic motor and sensory responses to various stimuli. Symptoms of increased stool frequency typically cease at night, alternate with periods of constipation, are accompanied by abdominal pain relieved with defecation, and rarely result in weight loss.

**Factitial Causes** Factitial diarrhea accounts for up to 15% of unexplained diarrheas referred to tertiary care centers. Either as a form of *Munchausen syndrome* (deception or self-injury for secondary gain) or *eating disorders*, some patients covertly self-administer laxatives alone or in combination with other medications (e.g., diuretics) or surreptitiously add water or urine to stool sent for analysis. Such patients are typically women, often with histories of psychiatric illness and disproportionately involved in careers in health care. Hypotension and hypokalemia are common co-presenting features. The evaluation of such patients may be difficult: contamination of the stool with water or urine is suggested by very low or high stool osmolarity, respectively. Such patients often deny this possibility when confronted, but they do benefit from psychiatric counseling when they acknowledge their behavior.

## APPROACH TO THE PATIENT

### Chronic Diarrhea

The laboratory tools available to evaluate the very common problem of chronic diarrhea are extensive, and many are costly and invasive. As such, the diagnostic evaluation must be rationally directed by a careful history, including medications, and physical examination (Fig. 49-4). When this strategy is unrevealing, simple triage tests are often warranted to direct the choice of more complex investigations

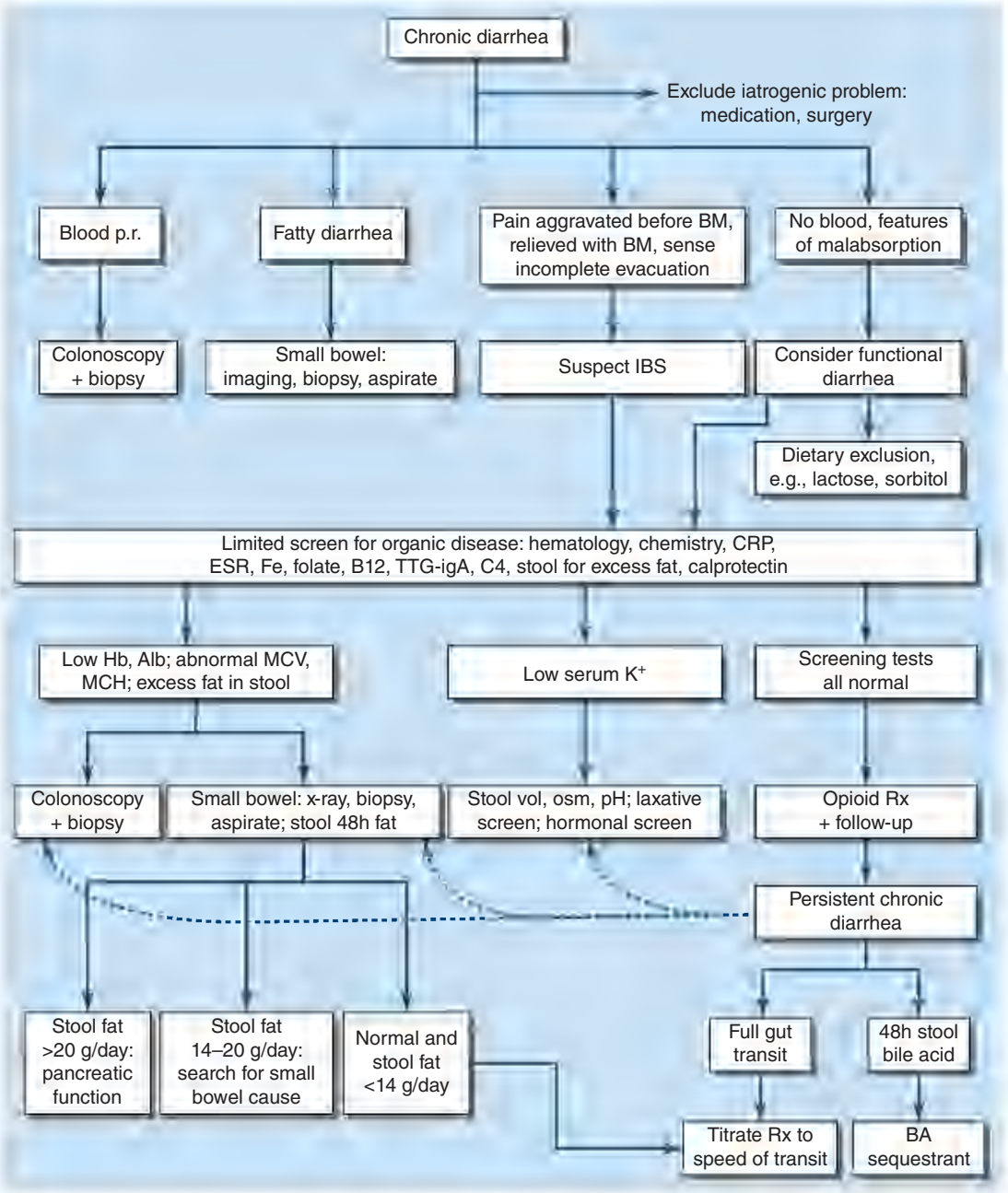


**FIGURE 49-4 Algorithm for management of chronic diarrhea.** Patients undergo an initial evaluation based on different symptom presentations, leading to selection of patients for imaging, biopsy analysis, and limited screens for organic diseases. Alb, albumin; BA, bile acid; BM, bowel movement; C4, 7 α-hydroxy-4-cholesten-3-one; CRP, C-reactive protein; ESR, erythrocyte sedimentation rate; Hb, hemoglobin; Hx, history; IBS, irritable bowel syndrome; MCH, mean corpuscular hemoglobin; MCV, mean corpuscular volume; osm, osmolality; p.r., per rectum; Rx, treatment; TTG, tissue transglutaminase. *(Reproduced with permission from M Camilleri, JH Sellin, KE Barrett: Pathophysiology, evaluation, and management of chronic watery diarrhea. Gastroenterology 152:515, 2017.)*

**TABLE 49-4 Physical Examination in Patients with Chronic Diarrhea**

1. Are there general features to suggest malabsorption or inflammatory bowel disease (IBD) such as anemia, arthritis, dermatitis herpetiformis, edema, or clubbing?
2. Are there features to suggest underlying autonomic neuropathy or collagen-vascular disease in the pupils, orthostasis, skin, hands, or joints?
3. Is there an abdominal mass or tenderness?
4. Are there any abnormalities of rectal mucosa, rectal defects, or altered anal sphincter functions?
5. Are there any mucocutaneous manifestations of systemic disease such as dermatitis herpetiformis (celiac disease), erythema nodosum (ulcerative colitis), flushing (carcinoid), or oral ulcers for IBD or celiac disease?

(Fig. 49-4). The history, physical examination **(Table 49-4)**, and routine blood studies should attempt to characterize the mechanism of diarrhea, identify diagnostically helpful associations, and assess the patient's fluid/electrolyte and nutritional status. Patients should be questioned about the onset, duration, pattern, aggravating (especially diet) and relieving factors, and stool characteristics of their diarrhea (such as relationship to food ingestion, presence of nocturnal episodes, undigested food or fat in the stool, and consistency based on Bristol Stool Form Scale). The presence or absence of fecal incontinence, fever, weight loss, pain, certain exposures (travel, medications, contacts with diarrhea), and common extraintestinal manifestations (skin changes, arthralgias, oral aphthous ulcers) should be noted. A family history of IBD or celiac disease may indicate those possibilities. Physical findings may offer clues such as a skin rash, thyroid mass, wheezing, heart murmurs, edema, hepatomegaly, abdominal masses, lymphadenopathy, mucocutaneous abnormalities, perianal fistulas, or anal sphincter laxity. Peripheral blood leukocytosis, elevated sedimentation rate, or C-reactive protein suggests inflammation; anemia reflects blood loss or nutritional deficiencies; or eosinophilia may occur with parasitoses, neoplasia, collagen-vascular disease, allergy, or eosinophilic gastroenteritis. Blood chemistries may demonstrate electrolyte, hepatic, or other metabolic disturbances. Measuring IgA tissue transglutaminase antibodies while eating gluten may help detect celiac disease. Bile acid diarrhea is confirmed by a scintigraphic radiolabeled bile acid retention test; where unavailable, a screening blood test (serum 7αC4 or FGF-19), measurement of fecal bile acids, or a therapeutic trial with a bile acid sequestrant (e.g., cholestyramine, colestipol or colesevelam) is indicated.

A therapeutic trial is often appropriate, definitive, and highly cost-effective when a specific diagnosis is suggested on the initial physician encounter. For example, chronic watery diarrhea, which ceases with fasting in an otherwise healthy young adult, may justify a trial of a lactose-restricted diet; bloating and diarrhea persisting since a mountain backpacking trip may warrant a trial of metronidazole for likely giardiasis; and postprandial diarrhea persisting following resection of terminal ileum might be due to bile acid malabsorption and be treated with cholestyramine, colestipol, or colesevelam before further evaluation. Persistent symptoms require additional investigation.

Certain diagnoses may be suggested on the initial encounter (e.g., idiopathic IBD); however, additional focused evaluations may be necessary to confirm the diagnosis and characterize the severity or extent of disease so that treatment can be best guided. Patients suspected of having IBS should be initially evaluated with flexible sigmoidoscopy with colorectal biopsies to exclude IBD, or particularly microscopic colitis, which is clinically indistinguishable from IBS with diarrhea or functional diarrhea. Patients with normal findings might be reassured and, as indicated, treated empirically with antispasmodics, antidiarrheals, or antidepressants (e.g., tricyclic agents). Any patient who presents with chronic diarrhea and hematochezia should be evaluated with stool microbiologic studies and colonoscopy.

In an estimated two-thirds of cases, the cause for chronic diarrhea remains unclear after the initial encounter, and further testing is required. Quantitative stool collection and analyses can yield important objective data that may establish a diagnosis or characterize the type of diarrhea as a triage for focused additional studies (Fig. 49-4). If stool weight is >200 g/d, additional stool analyses should be performed that might include electrolyte concentration, pH, occult blood testing, leukocyte inspection (or leukocyte protein assay), fat and bile acid quantitation, and laxative screens.

For secretory diarrheas (watery, with normal osmotic gap), possible medication-related side effects or surreptitious laxative use should be reconsidered. Stool microbiologic studies should be done with a multiplex panel initially or, if negative and diarrhea is persistent, fecal bacterial cultures (including media for *Aeromonas* and *Plesiomonas*), inspection for ova and parasites, and *Giardia* antigen assay (the most sensitive test for giardiasis). Small-bowel bacterial overgrowth can be excluded by intestinal aspirates with quantitative cultures or with glucose or lactulose breath tests involving measurement of breath hydrogen, methane, or other metabolite with an early peak in breath excretion. Note, interpretation of these breath tests may be confounded by disturbances of intestinal transit, e.g., if the gas peak is beyond 60 min. Upper endoscopy and colonoscopy with biopsies and small-bowel x-rays (formerly barium, but increasingly CT with enterography or magnetic resonance with enteroclysis) are helpful to rule out structural or occult inflammatory disease. When suggested by history or other findings, screens for peptide hormones should be pursued (e.g., serum gastrin, VIP, calcitonin, thyroid hormone/thyroid-stimulating hormone, urinary 5-hydroxyindolacetic acid, histamine).

Further evaluation of osmotic diarrhea should include tests for lactose and fructose intolerance and magnesium ingestion; intolerance of fructose is usually the result of ingestion of high fructose corn sweetener in which the fructose is ingested in the absence of sufficient glucose, which normally enhances fructose uptake. Low fecal pH suggests carbohydrate malabsorption; lactose malabsorption can be confirmed by lactose breath testing or by a therapeutic trial with lactose exclusion and observation of the effect of lactose challenge (e.g., a liter of milk). Lactase determination on small-bowel biopsy is not generally available. If fecal magnesium or laxative levels are elevated, inadvertent or surreptitious ingestion should be considered and psychiatric help should be sought.

For those with proven fatty diarrhea, endoscopy with small-bowel biopsy (including aspiration for quantitative cultures, if available) should be performed; if this procedure is unrevealing, a small-bowel radiograph is often an appropriate next step. If small-bowel studies are negative or if pancreatic disease is suspected, pancreatic exocrine insufficiency (PEI) should be excluded if possible with direct tests, such as the secretin-cholecystokinin stimulation test or a variation that could be performed endoscopically. In general, indirect tests for PEI such as assay of fecal elastase or chymotrypsin activity or a bentiromide test may be performed, but they have low sensitivity and specificity.

Chronic inflammatory-type diarrheas should be suspected by the presence of blood or leukocytes in the stool. Such findings warrant stool pathogen testing; colonoscopy with biopsies; and, if indicated, small-bowel imaging studies.

## TREATMENT

### Chronic Diarrhea

Treatment of chronic diarrhea depends on the specific etiology and may be curative, suppressive, or empirical. If the cause can be eradicated, treatment is curative as with resection of a colorectal cancer, antibiotic administration for Whipple's disease or tropical sprue, or discontinuation of a drug. For many chronic conditions, diarrhea can be controlled by suppression of the underlying mechanism. Examples include restriction of dietary lactose for lactase deficiency or gluten for celiac sprue, use of anti-inflammatory agents for idiopathic IBDs, bile acid sequestrants for bile acid malabsorption, PPIs

for the gastric acid hypersecretion of gastrinomas, somatostatin analogues such as octreotide for malignant carcinoid syndrome, prostaglandin inhibitors such as indomethacin for medullary carcinoma of the thyroid, and pancreatic enzyme replacement for pancreatic insufficiency. When the specific cause or mechanism of chronic diarrhea evades diagnosis, empirical therapy may be beneficial. Mild opiates, such as diphenoxylate or loperamide, are often helpful in mild or moderate watery diarrhea. For those with more severe diarrhea, codeine or tincture of opium may be beneficial. Such antimotility agents should be avoided with severe IBD because toxic megacolon may be precipitated. Clonidine, an $\alpha_2$-adrenergic agonist, may allow control of diabetic diarrhea, although the medication may be poorly tolerated because it causes postural hypotension; an alternative to oral administration is a clonidine skin patch. The 5-$HT_3$ receptor antagonists (e.g., alosetron, ondansetron) may relieve diarrhea and urgency in patients with IBS diarrhea. Other medications approved for the treatment of diarrhea associated with IBS are the nonabsorbed antibiotic, rifaximin, and the mixed μ-opioid receptor (OR) and κ-OR agonist and δ-OR antagonist, eluxadoline. The latter may induce sphincter of Oddi spasm and subsequent acute pancreatitis, usually in patients with prior cholecystectomy. Crofelemer, an antisecretory agent, has been used for AIDS-associated diarrhea. For all patients with chronic diarrhea, fluid and electrolyte repletion is an important component of management (see "Acute Diarrhea," earlier). Replacement of fat-soluble vitamins may also be necessary in patients with chronic steatorrhea.

## CONSTIPATION

### DEFINITION

Constipation is a common complaint in clinical practice and usually refers to persistent, difficult, infrequent, or seemingly incomplete defecation. Because of the wide range of normal bowel habits, constipation is difficult to define precisely. Most persons have at least three bowel movements per week; however, low stool frequency alone is not the sole criterion for the diagnosis of constipation. Many constipated patients have a normal frequency of defecation but complain of excessive straining, hard stools, lower abdominal fullness, or a sense of incomplete evacuation. The individual patient's symptoms must be analyzed in detail to ascertain what is meant by "constipation" or "difficulty" with defecation.

Stool form and consistency (which can be assessed using the Bristol Stool Form Scale) are well correlated with the time elapsed from the preceding defecation. Hard, pellety stools occur with slow transit, whereas loose, watery stools are associated with rapid transit. Both small pellety or very large stools are more difficult to expel than normal stools.

The perception of hard stools or excessive straining is more difficult to assess objectively, and the need for enemas or digital disimpaction is a clinically useful way to corroborate the patient's perceptions of difficult defecation.

Psychosocial or cultural factors may also be important. A person whose parents attached great importance to daily defecation will become greatly concerned when they miss a daily bowel movement; some children withhold stool to gain attention or because of fear of pain from anal irritation; and some adults habitually ignore or delay the call to have a bowel movement.

### CAUSES

Pathophysiologically, chronic constipation generally results from sedentary lifestyle, habitually ignoring the call to stool, inadequate fiber or fluid intake, or disordered colonic transit or anorectal function. These result from neurogastroenterologic disturbance, certain drugs, advancing age, or in association with a large number of systemic diseases that affect the GI tract (Table 49-5). Constipation of recent onset may be a symptom of significant organic disease such as colorectal cancer, anorectal irritation, or stricture. In *idiopathic constipation*, a subset of patients exhibits delayed emptying of the ascending and transverse colon with prolongation of transit (often in the proximal colon) and a reduced frequency of propulsive HAPCs. *Outlet obstruction to defecation* (also called *evacuation disorders or dyssynergic defecation*) accounts for at least a quarter of cases presenting with constipation in tertiary care and may cause delayed colonic transit, which is usually corrected by biofeedback retraining of the disordered defecation. Constipation of any cause may be exacerbated by hospitalization or chronic illnesses that lead to physical or mental impairment and result in inactivity or physical immobility.

**TABLE 49-5 Causes of Constipation in Adults**

| TYPES OF CONSTIPATION AND CAUSES | EXAMPLES |
|---|---|
| **Recent Onset** | |
| Colonic obstruction | Neoplasm; stricture: ischemic, diverticular, inflammatory |
| Anal sphincter spasm | Anal fissure, painful hemorrhoids |
| Medications | |
| **Chronic** | |
| Irritable bowel syndrome | Constipation-predominant, alternating |
| Medications | $Ca^{2+}$ blockers, antidepressants |
| Colonic pseudoobstruction | Slow-transit constipation, megacolon (rare Hirschsprung's, Chagas' diseases) |
| Disorders of rectal evacuation | Pelvic floor dysfunction; anismus; descending perineum syndrome; rectal mucosal prolapse; rectocele |
| Endocrinopathies | Hypothyroidism, hypercalcemia, pregnancy |
| Psychiatric disorders | Depression, eating disorders, drugs |
| Neurologic disease | Parkinsonism, multiple sclerosis, spinal cord injury |
| Generalized muscle disease | Progressive systemic sclerosis |

## APPROACH TO THE PATIENT

### Constipation

A careful history should explore the patient's symptoms and confirm whether they are indeed constipated based on frequency (e.g., fewer than three bowel movements per week), consistency (lumpy/hard), excessive straining, prolonged defecation time, or need to support the perineum or digitate the anorectum to facilitate stool evacuation. These latter items identified in the history suggest the presence of a rectal evacuation disorder. In the vast majority of cases (probably >90%), there is no underlying cause (e.g., cancer, depression, or hypothyroidism), and constipation responds to ample hydration, exercise, and supplementation of dietary fiber (15–25 g/d). A good diet and medication history and attention to psychosocial issues are key. Physical examination and, particularly, rectal examination are mandatory and should exclude fecal impaction and most of the important diseases that present with constipation and possibly indicate features suggesting an evacuation disorder (e.g., high anal sphincter tone, failure of perineal descent, or paradoxical puborectalis contraction or puborectalis tenderness during straining to simulate stool evacuation).

The presence of weight loss, rectal bleeding, or anemia with constipation mandates either flexible sigmoidoscopy plus barium enema or colonoscopy alone, to exclude structural diseases such as colorectal cancer (CRC) or strictures; however, screening should start earlier in patients with a known or suspected predisposing hereditary CRC syndrome and/or a family history of CRC. Colonoscopy alone is most cost-effective in this setting because it provides an opportunity to biopsy mucosal lesions, perform polypectomy, or dilate strictures. Barium enema has advantages over colonoscopy in the patient with isolated constipation because it is less costly and identifies colonic dilation and all significant mucosal lesions or strictures that are likely to present with constipation. Melanosis coli, or pigmentation of the colon mucosa, indicates the use of anthraquinone laxatives such as cascara or senna. An unexpected

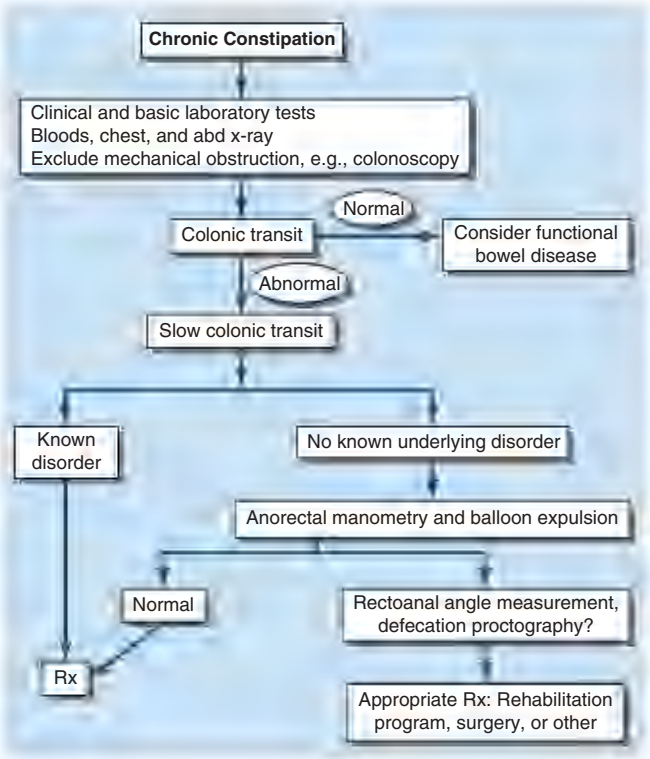


**FIGURE 49-5 Algorithm for the management of constipation.** abd, abdominal; Rx, treatment.

disorder such as megacolon or cathartic colon may also be detected by colonic radiographs. Measurement of serum calcium, potassium, and thyroid-stimulating hormone levels will identify rare patients with metabolic disorders.

Patients with more troublesome constipation may not respond to fiber alone and may be helped by a bowel-training regimen, which involves taking an osmotic laxative (e.g., polyethylene glycol magnesium salts, lactulose, sorbitol) and evacuating with enema or suppository (e.g., glycerin or bisacodyl) as needed. After breakfast, a distraction-free 15–20 min on the toilet without straining is encouraged. The use of squat position may help. Excessive straining may lead to development of hemorrhoids and, if there is weakness of the pelvic floor or injury to the pudendal nerve, may result in obstructed defecation from descending perineum syndrome several years later. Those few who do not benefit from the simple measures delineated above or require long-term treatment or fail to respond to potent laxatives should undergo further investigation **(Fig. 49-5)**. Novel agents that induce secretion (e.g., lubiprostone, a chloride channel activator, or linaclotide or plecanatide, both guanylate cyclase C agonists that activate chloride secretion, or tenapanor, an NHE3 inhibitor) are also available.

## INVESTIGATION OF SEVERE CONSTIPATION

A small minority (probably <5%) of patients have severe or "intractable" constipation; at least 25% (of patients seen in gastroenterology clinics) have evacuation disorders. These are the patients most likely to require evaluation by gastroenterologists or in referral centers. Further observation of the patient may occasionally reveal a previously unrecognized cause, such as an evacuation disorder, laxative abuse, malingering, or psychological disorder. In these patients, evaluations of the physiologic function of the colon and pelvic floor and of psychological status aid in the rational choice of treatment. Even among these highly selected patients with severe constipation, a cause can be identified in only about one-third of tertiary referral patients, with the others being diagnosed with normal transit constipation. Since evacuation disorders also retard colonic transit through the left colon or the entire colon, anorectal and pelvic floor testing should precede transit measurements if there is clinical suspicion of an evacuation disorder. If an evacuation disorder is identified on testing, colonic transit may be unnecessary.

**Measurement of Colonic Transit** Radiopaque marker transit tests are easy, repeatable, generally safe, inexpensive, reliable, and highly applicable in evaluating constipated patients in clinical practice. Several validated methods are very simple. For example, radiopaque markers are ingested; an abdominal flat film taken 5 days later should indicate passage of 80% of the markers out of the colon without the use of laxatives or enemas. This test does not provide useful information about the transit profile of the stomach and small bowel. An alternative approach involves ingestion of 24 radiopaque markers on 3 successive days and an abdominal radiograph on the fourth day. The number of markers counted in the radiograph is an estimate of the colonic transit in hours. The collection of gas in the rectum between the level of the ischial spines and the lower border of the sacroiliac joints may suggest the presence of a rectal evacuation disorder as the cause of constipation.

Radioscintigraphy has been used to noninvasively characterize normal, accelerated, or delayed colonic function over 24–72 h with low radiation exposure. These approaches simultaneously assess gastric, small bowel (which may be important in ~20% of patients with delayed colonic transit because they reflect a more generalized GI motility disorder), and colonic transit. The disadvantages are the greater cost and the need for specific materials prepared in a nuclear medicine laboratory. Newer approaches consisting of extracorporeal detector plate, ingestible electromagnetic capsules, and analysis software are becoming available and may obviate need for approaches involving radiation.

**Anorectal and Pelvic Floor Tests** Pelvic floor dysfunction is suggested by the inability to completely evacuate the rectum, a feeling of persistent rectal fullness, rectal pain, the need to extract stool from the rectum digitally, application of pressure on the posterior wall of the vagina (suggestive of anterior rectocele), support of the perineum during straining, and excessive straining. These significant symptoms contrast with simple sense of incomplete rectal evacuation, which is common in IBS.

Formal psychological evaluation may identify eating disorders including ARFID (avoidant/restrictive food intake disorder), "control issues," depression, or posttraumatic stress disorders that may respond to cognitive or other intervention and may be important in restoring quality of life to patients who might present with chronic constipation.

A simple clinical test in the office to document a nonrelaxing puborectalis muscle is to have the patient strain to expel the index finger during a digital rectal examination. Motion of the puborectalis posteriorly with descent during straining indicates proper coordination of the pelvic floor muscles. Motion anteriorly with paradoxical puborectalis contraction or limited perineal descent (<1.5 cm) during simulated evacuation indicates pelvic floor dysfunction.

Measurement of perineal descent is relatively easy to gauge clinically by placing the patient in the left decubitus position and watching the perineum to detect inadequate descent (<1.5 cm, a sign of pelvic floor dysfunction) or perineal ballooning during straining relative to bony landmarks (>4 cm, suggesting excessive perineal descent).

A useful overall test of evacuation is the balloon expulsion test. A balloon-tipped urinary catheter is placed and inflated with 50 mL of water. Normally, a patient can expel it while seated on a toilet or in the left lateral decubitus position. In the lateral position, the weight needed to facilitate expulsion of the balloon is determined; normally, expulsion occurs unaided within 1 min.

Anorectal manometry, when used in the evaluation of patients with severe constipation, may find an excessively high resting (>80 mmHg) or squeeze anal sphincter tone, suggesting anismus (anal sphincter spasm). The rectoanal pressure difference along with a balloon expulsion time >60 s is consistently identified as the most useful indicators of evacuation disorders. This test also identifies rare syndromes, such as Hirschsprung's disease, by absence of the rectoanal inhibitory reflex (RAIR). However, the RAIR may be absent in patients with enlarged rectum due to prolonged retention of stool.

Defecography (a dynamic barium enema including lateral views obtained during barium expulsion or a magnetic resonance defecogram) measures changes in rectoanal angle, degree of pelvic floor relaxation and perineal descent, as well as anatomic defects of the rectum such as internal mucosal prolapse, intussusception, and enteroceles or rectoceles. Surgically remediable conditions are identified in only a few patients with severe, whole-thickness intussusception or extremely large rectocele that fills preferentially during attempts at defecation instead of expulsion through the anus. Magnetic resonance defecography provides more information about the structure of the pelvic floor, distal colorectum, and anal sphincters and some information about the function during defecatory effort; however, it is performed in supine rather than seated position.

Neurologic testing (electromyography) is rarely performed in the evaluation of patients with constipation. The absence of neurologic signs in the lower extremities suggests that any documented denervation of the puborectalis results from pelvic (e.g., obstetric) injury or from stretching of the pudendal nerve by chronic, long-standing straining. Constipation is common among patients with spinal cord injuries, neurologic diseases such as Parkinson's disease, multiple sclerosis, and diabetic neuropathy.

Spinal-evoked responses during electrical rectal stimulation or stimulation of external anal sphincter contraction by applying magnetic stimulation over the lumbosacral cord identify patients with limited sacral neuropathies with sufficient residual nerve conduction to attempt biofeedback training.

In summary, a balloon expulsion test is an important component of anorectal manometry as screening tests for anorectal dysfunction. An anatomic evaluation of the rectum or anal sphincters is rarely required when evaluating patients in whom obstructed defecation is suspected. Indications for anatomic evaluation are symptoms of rectal mucosal prolapse, pressure of the posterior wall of the vagina to facilitate defecation (suggestive of anterior rectocele), or prior pelvic surgery that may be complicated by enterocele.

## TREATMENT

### Constipation

After the cause of constipation is characterized, a treatment decision can be made. Slow-transit constipation requires aggressive medical or surgical treatment; conversely, anismus or pelvic floor dysfunction usually responds to biofeedback management (Fig. 49-5). The remaining ~60% of patients with constipation have normal colonic transit and can be treated symptomatically. Patients with spinal cord injuries or other neurologic disorders require a dedicated bowel regimen that often includes rectal stimulation, enema therapy, and carefully timed laxative therapy.

Patients with constipation are treated with bulk (fiber, psyllium), osmotic (milk of magnesia, lactulose, polyethylene glycol), secretory (linaclotide, lubiprostone, plecanatide, tenapanor), and prokinetic or stimulant laxatives (including diphenyl methanes such as bisacodyl and sodium picosulfate, senna alkaloids, and the 5-$HT_4$ agonist prucalopride). If a 3- to 6-month trial of medical therapies fails, unassociated with obstructed defecation, the patient should be considered for laparoscopic colectomy with ileorectostomy; however, this should not be undertaken for pain or in the presence of continued evidence of an evacuation disorder or a generalized GI dysmotility. Referral to a specialized center for further tests of colonic motor function is warranted. The decision to resort to surgery is facilitated by the presence of megacolon and megarectum or documented sigmoid or cecal volvulus. The complications after surgery include small-bowel obstruction (11%) and fecal soiling, particularly at night during the first postoperative year. Frequency of defecation is 3–8 per day during the first year, dropping to 1–3 per day from the second year after surgery.

Patients who have a combined (evacuation and transit/motility) disorder should first pursue pelvic floor retraining (biofeedback and muscle relaxation), psychological counseling, and dietetic advice. If symptoms are intractable despite biofeedback and optimized medical therapy, colectomy and ileorectostomy could be considered as long as the evacuation disorder is resolved and optimized medical therapy is unsuccessful. In patients with pelvic floor dysfunction alone, biofeedback training for 3 months has a 70–80% success rate, measured by the acquisition of comfortable stool habits. Attempts to manage pelvic floor dysfunction with operations (internal anal sphincter or puborectalis muscle division) or injections with botulinum toxin have achieved only mediocre success and have been largely abandoned. Persistence of the evacuation disorder despite physical therapy with biofeedback may require regular enemas or consideration of a loop- or end-ileostomy to relieve the intractable constipation.

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# 50 Unintentional Weight Loss

J. Larry Jameson

Involuntary or unintentional weight loss (UWL) is frequently insidious and can have important implications, often serving as a harbinger of serious underlying disease. Clinically important weight loss is defined as the loss of >5% of body weight over a period of 6–12 months. UWL is not uncommon in individuals aged ≥65 years. It can be challenging to recognize in patients with preexisting obesity or inadequate